

Itô's Lemma

Ordinary calculus has the *chain rule*: how a function of a changing quantity changes. Itô's lemma is the chain rule for functions of Brownian motion. It has one extra term, and this chapter shows exactly where that term comes from.

4.1 The Surviving Second-Order Term

Take a function f , such as e^x , and follow $f(W_t)$ along a Brownian path. Over one small piece of time, the change in f can be split into a sum of simpler terms: a first-order term using the slope f' , a second-order term using the curvature f'' , and so on. Add each kind of term up over all n pieces. The bars show the totals along one path.

Predict first: As the pieces shrink, which total will NOT shrink to 0 (apart from the first-order one)?

- The second-order total
- The third-order total
- Both will shrink to 0

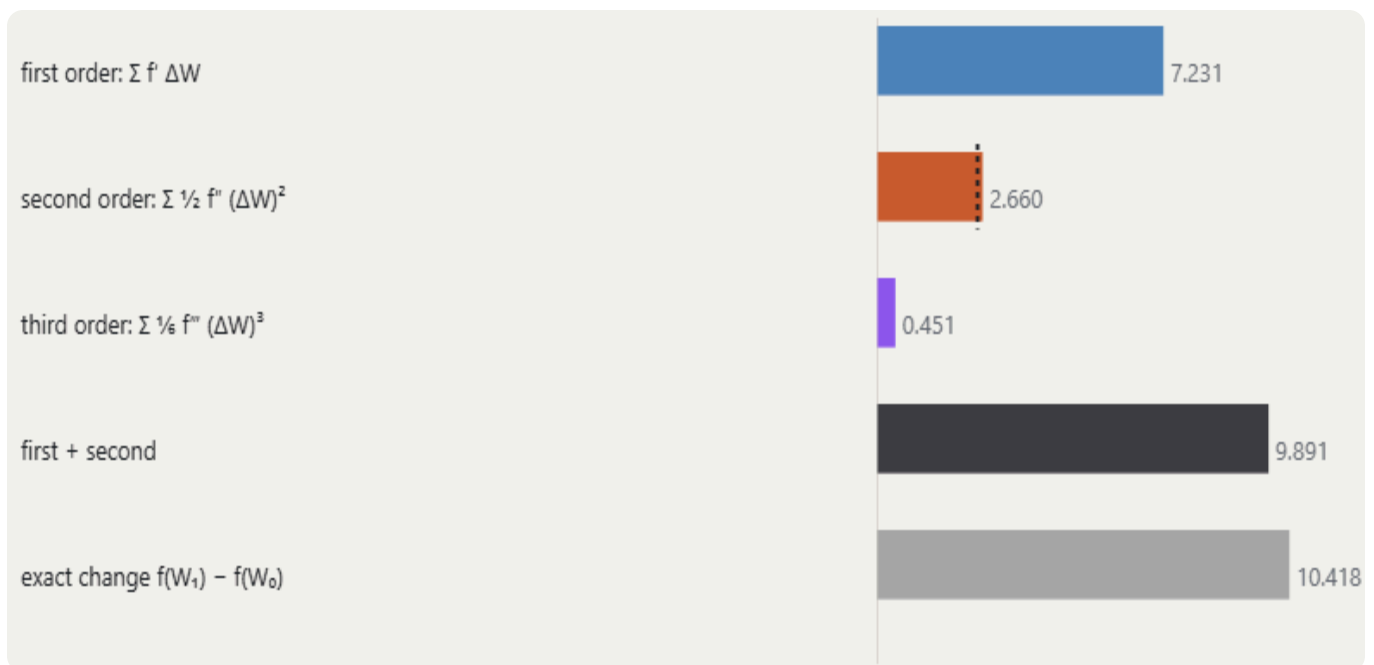
Answer: "The second-order total". Squared rises add up to elapsed time rather than 0 (section 2.1), so the second-order total settles to $\frac{1}{2}f''(W)dt$. Cubed rises are too small and vanish.

Drag n up. In ordinary calculus, every term beyond the first would shrink to nothing. Watch which ones do here.

$$f(W_{t+\Delta t}) - f(W_t) = f'(W_t) \Delta W + \frac{1}{2}f''(W_t) (\Delta W)^2 + \frac{1}{6}f'''(W_t) (\Delta W)^3 + \dots$$

f	Any smooth function, chosen below: e^x , $\sin x$ or x^3 .
f', f'', f'''	Its first, second and third <i>derivatives</i> . f' is the slope, f'' is how fast the slope changes (curvature), and f''' is how fast the curvature changes. For x^3 : $f' = 3x^2$, $f'' = 6x$, $f''' = 6$.
$\Delta t, \Delta W$	The length of one small piece of time, and the path's rise across it.
$\dots$	Higher-order terms, which use fourth and higher derivatives. They shrink even faster than the third-order term.
Taylor expansion	The name of this recipe: a change in f is the slope $\times$ the step, plus a curvature correction $\times$ the step squared, and so on.

In plain words: for a smooth curve, a step of size Δt makes the squared term about $(\Delta t)^2$. Summed over $n = 1/\Delta t$ pieces, that totals about Δt , which vanishes. For Brownian motion the step is $\Delta W \approx \sqrt{\Delta t}$, so $(\Delta W)^2 \approx \Delta t$. Summed over all the pieces, that is a fixed amount that does *not* vanish: section 2.1's quadratic variation again. The cubed term is about $(\Delta t)^{3/2}$ per piece and does vanish in total. So exactly one extra term survives: the second-order one.



$f(x) = e^x$, $n = 8$ pieces. Second-order total 2.660 is heading for $\frac{1}{2} \int f''(W) dt = 2.531$ (dashed tick), not 0. The remaining gap is random and shrinks roughly like $1/\sqrt{n}$. Third-order total 0.4511 shrinks toward 0. First + second = 9.891 vs exact 10.418.

As the pieces shrink, the third-order bar disappears. The second-order bar settles to $\frac{1}{2} \int_0^1 f''(W_t) dt$ (the dashed tick), and "first + second" matches the exact change. Replacing

$(\Delta W)^2$ by Δt , the rule $(dW)^2 = dt$, is all it takes to turn the second-order term into an ordinary time integral.

Going deeper: the precise statement

Sketch of the proof of Itô's lemma. Write $f(W_t) - f(W_0) = \sum_i [f(W_{t_{i+1}}) - f(W_{t_i})]$ and expand each term with Taylor's theorem. Three facts finish the proof as the partition is refined:

- $\sum_i f'(W_{t_i}) \Delta W_i \rightarrow \int_0^t f'(W_s) dW_s$ by the definition of the Itô integral.
- $\sum_i f''(W_{t_i}) (\Delta W_i)^2 \rightarrow \int_0^t f''(W_s) ds$, a weighted version of quadratic variation (section 2.1).
- The remainder is at most a constant $\times \max_i |\Delta W_i| \cdot \sum_i (\Delta W_i)^2 \rightarrow 0 \cdot t = 0$.

Reference: Shreve, *Stochastic Calculus for Finance II*, §4.4.

Show the code (Python + NumPy)

```
import numpy as np
rng = np.random.default_rng(10)
N = 2 ** 16
W = np.concatenate([[0], np.cumsum(rng.normal(0, np.sqrt(1 / N), N))])
f = d1 = d2 = d3 = np.exp # f(x) = e^x is its own derivative
for n in (8, 2 ** 16):
    x, dW = W[:-1:N // n], np.diff(W[:-1:N // n])
    t1, t2, t3 = np.sum(d1(x) * dW), np.sum(d2(x) * dW ** 2) / 2, np.sum(d3(x) * dW ** 3) / 6
    print(f"n = {n:>5}: first {t1:.3f} second {t2:.3f} third {t3:+.4f} "
          f"first+second {t1 + t2:.3f} exact {f(W[-1]) - f(0):.3f}")
print(f"half the integral of f'' dt = {np.mean(d2(W[:-1])) / 2:.3f}")
```

Output when we ran it:

```
n =      8: first -1.026 second 0.435 third -0.0908 first+second -0.592 exact -0.668
n = 65536: first -0.937 second 0.269 third -0.0000 first+second -0.668 exact -0.668
half the integral of f'' dt = 0.267
```

4.2 Itô's Lemma in Action

Section 4.1 gives the new chain rule. The chart tests it. Along one Brownian path, the thick grey curve is the true $f(W_t)$. The two thin curves rebuild it from scratch by adding up small changes: orange uses Itô's lemma, purple uses the ordinary chain rule. Switch between functions.

Predict first: For $f(x) = x^2$, where will the purple naive chain-rule curve end up compared with the truth?

On top of it

Above it

Below it

Answer: "Below it". The naive rule misses $+\frac{1}{2}f'' dt = +dt$ at every step, which adds up to $+1$ by $t = 1$. So the naive curve ends about 1 below the true W_1^2 .

$$df(W_t) = f'(W_t) dW_t + \frac{1}{2}f''(W_t) dt$$

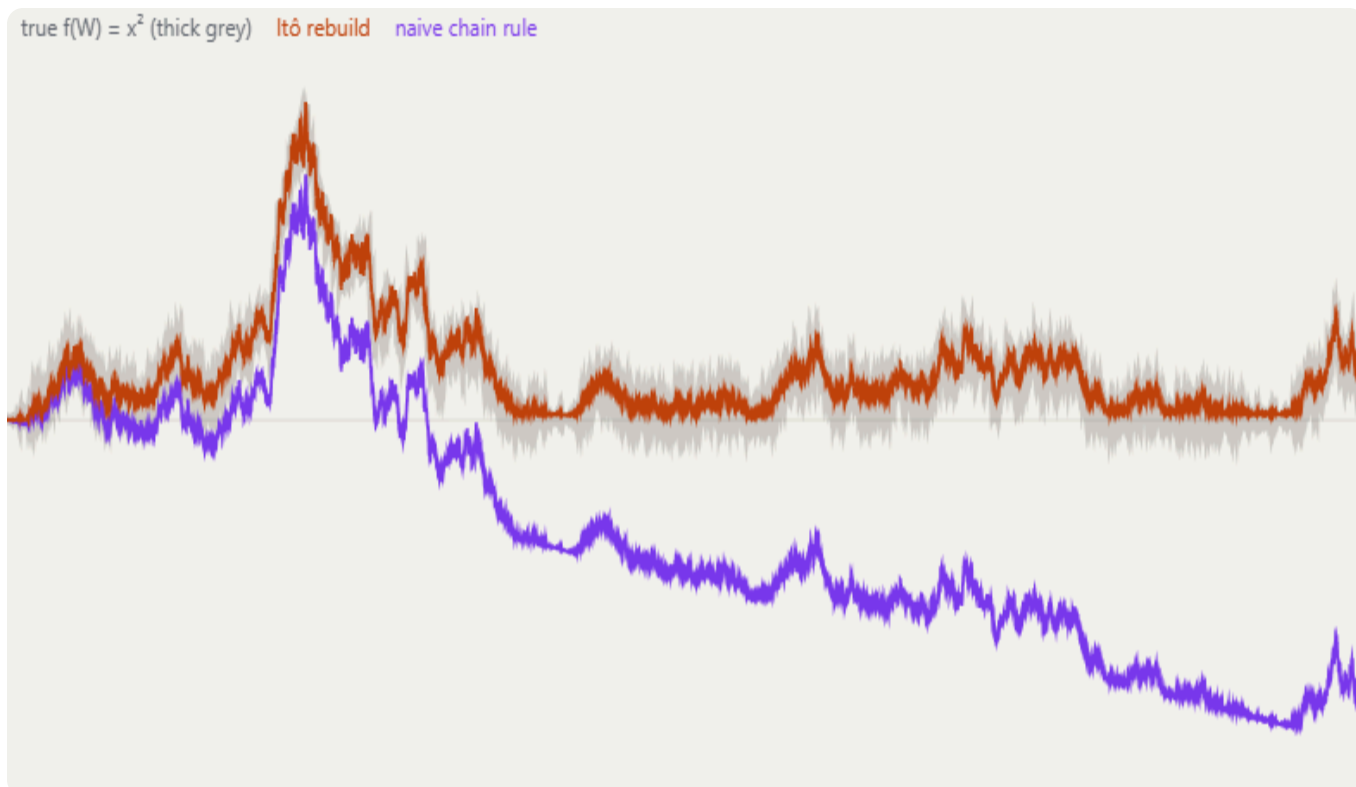
d "A tiny change in". $df(W_t)$ is the tiny change in $f(W_t)$ over a tiny time step dt , and dW_t is the tiny rise of the path over it. Adding up these tiny changes means taking an integral.

$f'(W_t) dW_t$ The ordinary chain-rule term: slope $\times$ change. Added up, it is an Itô integral (Chapter 3).

$\frac{1}{2}f''(W_t) dt$ The extra Itô term: half the curvature $\times$ the time step. Added up, it is an ordinary integral over time.

chain rule Ordinary calculus's rule $df(x) = f'(x) dx$: the change in f is its slope $\times$ the change in x . It is correct for smooth paths, and it misses the second term for Brownian ones.

In plain words: when f curves upward ($f'' > 0$), jiggling back and forth doesn't cancel out. A step up raises f by more than an equal step down lowers it. So $f(W)$ creeps upward on average, at rate $\frac{1}{2}f''$. The ordinary chain rule can't see this because it assumes paths are smooth. For $f(x) = x^2$, Itô's lemma says $d(W^2) = 2W dW + dt$. The " $+ dt$ " is the upward drift of W^2 you saw in section 2.2.



At $t = 1$: true value 0.129, Itô rebuild 0.147, naive chain rule -0.853. The naive rule is off by 0.982: the missing $\frac{1}{2} \int f''(W) dt = 1.000$, up to a small error from using 16,384 finite steps.

Itô's rebuild lies right on top of the truth, while the naive chain rule drifts away by exactly $\frac{1}{2} \int f''(W) dt$. This is **Itô's lemma**, the central tool of stochastic calculus. Everything that follows (solving random equations, pricing options) is Itô's lemma applied to the right function.

Going deeper: the precise statement

Itô–Doebelin formula, general form. If $dX_t = a_t dt + b_t dW_t$ and $f(t, x)$ is smooth, then

$$df(t, X_t) = \left(\partial_t f + a_t \partial_x f + \frac{1}{2} b_t^2 \partial_{xx} f \right) dt + b_t \partial_x f dW_t.$$

The bookkeeping rules are $dt \cdot dt = 0$, $dt \cdot dW = 0$ and $dW \cdot dW = dt$.

History. Kiyosi Itô published the formula in 1951. Wolfgang Doeblin had derived it independently in 1940, in a sealed envelope sent to the French Academy of Sciences shortly before his death in the war. The envelope was opened only in 2000, which is why the formula is increasingly called Itô–Doebelin.

References: Shreve, *Stochastic Calculus for Finance II*, §4.4. Øksendal, *Stochastic Differential Equations*, ch. 4.

Show the code (Python + NumPy)

```

import numpy as np
rng = np.random.default_rng(11)
steps = 16_384
dt = 1 / steps
W = np.concatenate([[0], np.cumsum(rng.normal(0, np.sqrt(dt), steps))])
x, dW = W[:-1], np.diff(W)
naive = np.sum(2 * x * dW)           # chain rule for f(x) = x^2: df = 2x dx
ito = naive + steps * dt             # Ito adds 1/2 f''(x) dt = dt at every step
print(f"true W_1^2 = {W[-1] ** 2:.3f}   Ito = {ito:.3f}   naive = {naive:.3f}")

```

Output when we ran it:

```

true W_1^2 = 0.577   Ito = 0.576   naive = -0.424

```

4.3 Itô vs Stratonovich

Section 3.1 showed two popular reading rules: Itô (left end) and Stratonovich (midpoint). They give different integrals, so the "same" equation means different things under each. Here, the same Brownian path drives the equation " X changes by $X \times$ the rise of W ", read both ways. Then tick the box to add a correction term to the Itô version.

Predict first: Before you tick the box: which curve will end higher?

☐ Itô

☐ Stratonovich

☐ They will coincide

Answer: "Stratonovich". Stratonovich solves to $e^{(W_t)}$, Itô to $e^{(W_t - t/2)}$: the same path, but Itô is lower by the factor $e^{(-t/2)}$. The conversion drift restores it.

$$dX_t = X_t \circ dW_t \iff dX_t = \frac{1}{2}X_t dt + X_t dW_t$$

X_t A quantity that grows or shrinks in proportion to itself, starting at $X_0 = 1$: a price moving in percentage terms, for example.

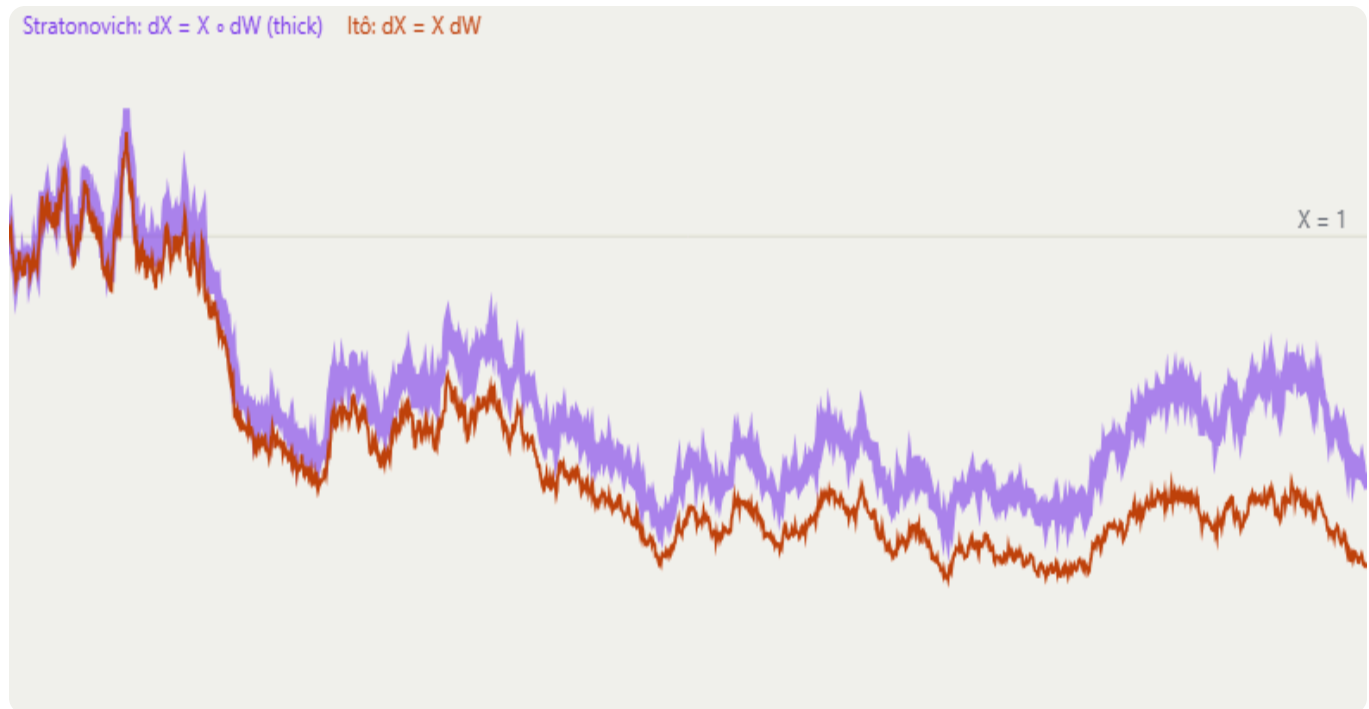
○ dW_t The circle marks the Stratonovich reading: X is read at the midpoint of each small step, like $\theta = \frac{1}{2}$ in section 3.1.

dW_t (no circle) The Itô reading: X is read at the start of each step.

$\iff$ "Describes exactly the same process as".

$\frac{1}{2}X_t dt$ The *conversion drift*: the amount you must add to an Itô equation to make it match the Stratonovich one.

In plain words: reading X at the midpoint means it has already reacted to half of the current step's rise, which quietly adds a little growth each step. On average that is $\frac{1}{2}X dt$, the same "peeking" effect as in section 3.1. So the Stratonovich equation grows like e^{W_t} , while the Itô equation without the correction grows like $e^{W_t - t/2}$. Add the drift to the Itô version and the two become identical. Neither reading is wrong. They are two languages, and the drift term translates between them.



At $t = 1$: Stratonovich 0.474 (exact $e^W = 0.474$); Itô 0.285 (exact $e^{(W - t/2)} = 0.288$).

Same noise, same-looking equation, different answers, until the conversion drift is added. Physicists often prefer Stratonovich, because the ordinary chain rule works for it. Finance uses Itô, because you can't trade on a price you haven't seen yet. The rest of this site uses Itô.

Going deeper: the precise statement

General conversion rule. For a smooth coefficient $b(x)$:

$$dX = a(X) dt + b(X) \circ dW \iff dX = \left(a(X) + \frac{1}{2} b(X) b'(X)\right) dt + b(X) dW.$$

In this section $a = 0$ and $b(x) = x$, so the extra drift is $\frac{1}{2}x$.

Why physicists like Stratonovich: the Wong–Zakai theorem. If you replace W by a smooth approximation (real noise always has some tiny correlation time), solve the ordinary differential equation, and then let the approximation become rough, the solutions converge to the *Stratonovich* solution. Stratonovich calculus also obeys the ordinary chain rule.

References: E. Wong & M. Zakai, *Ann. Math. Statist.* 36 (1965), 1560–1564. Øksendal, *Stochastic Differential Equations*, ch. 3.

Show the code (Python + NumPy)

```
import numpy as np
rng = np.random.default_rng(12)
steps = 2_000
dt = 1 / steps
dW = rng.normal(0, np.sqrt(dt), steps)
ito, ito_drift, strat = 1.0, 1.0, 1.0
for d in dW:
    ito += ito * d # dX = X dW, read at the left end
    ito_drift += 0.5 * ito_drift * dt + ito_drift * d # the same plus the conversion
drift
    guess = strat + strat * d # Stratonovich: Heun (midpoint) step
    strat += 0.5 * (strat + guess) * d
W1 = dW.sum()
print(f"Ito {ito:.3f} (e^(W - 1/2) = {np.exp(W1 - 0.5):.3f})    Ito + drift {ito_drift:.3f}"
      f"Stratonovich {strat:.3f} (e^W = {np.exp(W1):.3f})")
```

Output when we ran it:

```
Ito 1.027 (e^(W - 1/2) = 1.036)    Ito + drift 1.693    Stratonovich 1.708 (e^W = 1.708)
```

4.4 Two Brownian Motions

Real models rarely have just one source of randomness. Two stock prices, or a price and its volatility, are driven by two Brownian motions that tend to move together (or against each

other) without being identical. The slider ρ ("rho") sets how strongly the blue and purple paths move together.

Predict first: Set $\rho = 0.8$. The running sum of (rise of W_1) $\times$ (rise of W_2) will...

Settle at 0

Settle near 0.8

Grow forever

Answer: "Settle near 0.8". Each product of rises averages $\rho \times$ (time step), so the running sum settles on the line ρt : here 0.8 at $t = 1$. It is the two-path version of quadratic variation.

Drag ρ from -1 to 1 . Watch the paths, their shape in the plane, and the running sum at the bottom.

$$dW_t^{(1)} dW_t^{(2)} = \rho dt \quad \implies \quad d(XY) = X dY + Y dX + dX dY$$

$W^{(1)}, W^{(2)}$ Two Brownian motions, blue and purple. Each one on its own is an ordinary Brownian motion (Chapter 1).

ρ Their *correlation*, from -1 to 1 (section 1.2): 1 means they move in lockstep, 0 means independently, -1 means they are mirror images.

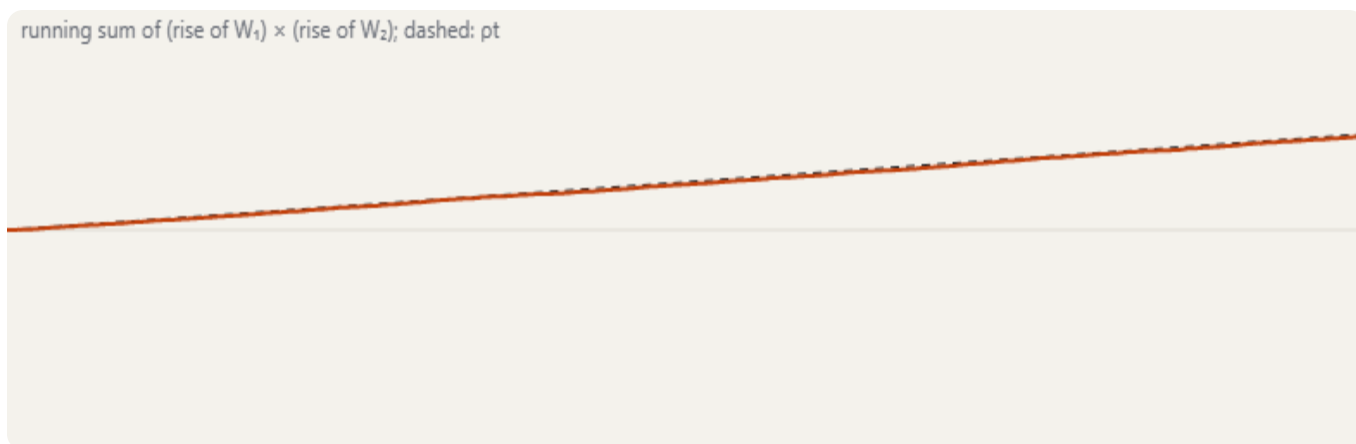
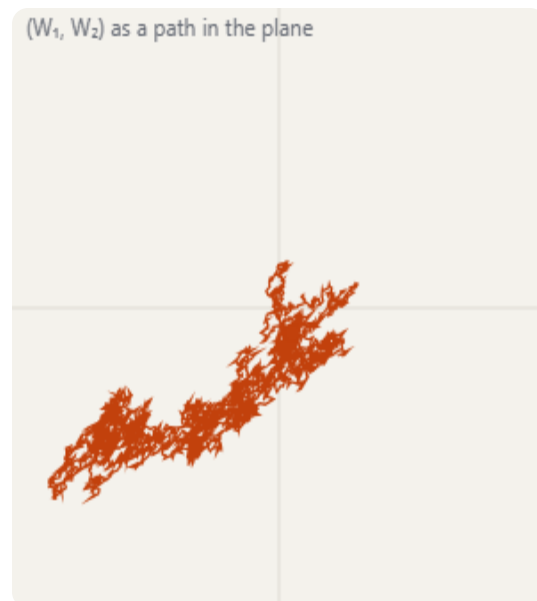
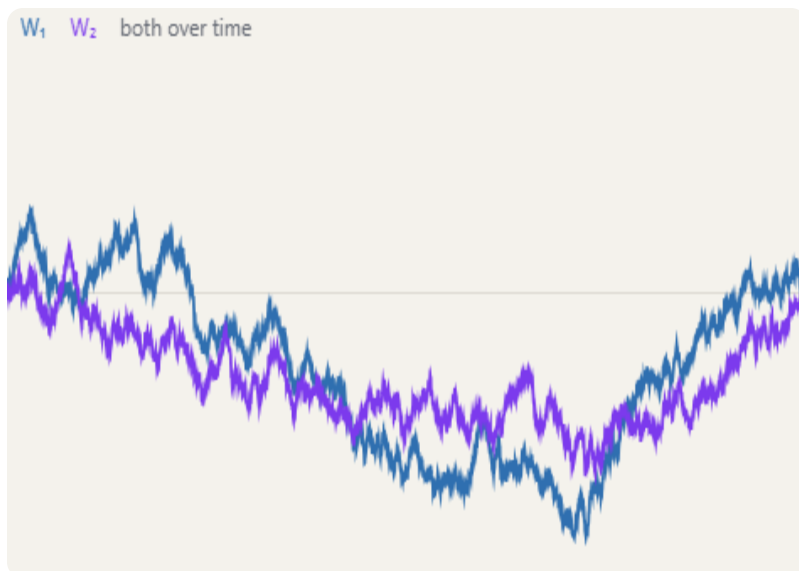
$dW^{(1)} dW^{(2)} = \rho dt$ Multiplying their tiny rises and adding up gives $\rho \times$ elapsed time. With $\rho = 1$ this is section 2.1's $(dW)^2 = dt$, and with $\rho = 0$ the products cancel out to 0 .

X, Y Any two processes built from these Brownian motions. Here simply $X = W^{(1)}$ and $Y = W^{(2)}$.

$d(XY)$ The tiny change in the product $X \times Y$.

$dX dY$ The extra term of the *Itô product rule*. In ordinary calculus it vanishes. Here it equals ρdt .

In plain words: you build a pair with correlation ρ by mixing: $W^{(2)}$ takes a share ρ of $W^{(1)}$'s kicks plus a share $\sqrt{1 - \rho^2}$ of its own independent kicks. Their rises then multiply to ρdt on average, and just as in section 2.1 the sum of those products is not random at all: it is exactly ρt . That is why the ordinary product rule $d(XY) = X dY + Y dX$ gains a third term, $dX dY$. It is Itô's lemma for a function of two variables.



Top right: the point (W_1, W_2) wanders in the plane. With ρ near ± 1 it is squeezed onto a diagonal, with $\rho = 0$ it spreads out equally in every direction. Bottom: the running sum settles on the dashed line ρt .

Sum of (rise of $W_1 \times$ rise of W_2) up to $t = 1$: 0.583 (theory: $\rho = 0.60$). Product rule at $t = 1$: $W_1 W_2 = -0.009$.
 $\int W_1 dW_2 + \int W_2 dW_1 = -0.592$, plus the extra $\rho t = 0.008$.

Two correlated Brownian motions have a *cross-variation* ρt , and it adds the extra ρdt to the product rule. With this, Itô's lemma extends to any number of Brownian motions. That is what pricing an option on two assets, or a stock whose volatility is itself random (the Heston model), requires.

Going deeper: the precise statement

Construction. If $B^{(1)}, B^{(2)}$ are independent Brownian motions, then $W^{(1)} = B^{(1)}$ and $W^{(2)} = \rho B^{(1)} + \sqrt{1 - \rho^2} B^{(2)}$ are Brownian motions with $\text{Cov}(W_t^{(1)}, W_t^{(2)}) = \rho t$ and cross-variation $[W^{(1)}, W^{(2)}]_t = \rho t$. In d dimensions, a correlation matrix C is factored as $C = LL^\top$ (the Cholesky factorisation) and one sets $W = LB$.

Multivariable Itô formula. For smooth $f(x, y)$:

$$df(X, Y) = f_x dX + f_y dY + \frac{1}{2} f_{xx} (dX)^2 + f_{xy} dX dY + \frac{1}{2} f_{yy} (dY)^2,$$

using $dW^{(i)} dW^{(j)} = \rho_{ij} dt$ and $dt \cdot dW = dt \cdot dt = 0$. The product rule is the case $f(x, y) = xy$.

Lévy's characterisation, two dimensions. Two continuous martingales starting at 0 with $[M^{(1)}]_t = [M^{(2)}]_t = t$ and $[M^{(1)}, M^{(2)}]_t = 0$ are independent Brownian motions.

Reference: Shreve, *Stochastic Calculus for Finance II*, §4.6.

Show the code (Python + NumPy)

```
import numpy as np
rng = np.random.default_rng(19)
rho, steps = 0.6, 100_000
dt = 1 / steps
dB1, dB2 = rng.normal(0, np.sqrt(dt), (2, steps))    # two independent noises
dW1 = dB1
dW2 = rho * dB1 + np.sqrt(1 - rho ** 2) * dB2        # correlated with dW1
W1 = np.concatenate([[0], np.cumsum(dW1)])
W2 = np.concatenate([[0], np.cumsum(dW2)])
print(f"sum of dW1 * dW2 = {np.sum(dW1 * dW2):.3f}    (theory: rho = {rho})")
integrals = np.sum(W1[:-1] * dW2) + np.sum(W2[:-1] * dW1)    # Ito integrals, left-end rule
print(f"W1 * W2 = {W1[-1] * W2[-1]:.3f}    integrals = {integrals:.3f}    integrals + rho = {integrals + rho:.3f}")
```

Output when we ran it:

```
sum of dW1 * dW2 = 0.598    (theory: rho = 0.6)
W1 * W2 = -0.063    integrals = -0.660    integrals + rho = -0.060
```

Chapter recap

Quick questions on this chapter's big ideas. Each answer is explained as soon as you choose.

1. By Itô's lemma, the tiny change $d(W^2)$ equals...

2W dW

2W dW + dt

2W dW - dt

Answer: "2W dW + dt". $f(x) = x^2$ has $f' = 2x$ and $f'' = 2$, so $df = 2W dW + \frac{1}{2} \cdot 2 dt = 2W dW + dt$.

2. Why does a second-order term survive in stochastic calculus but not in ordinary calculus?

$(dW)^2$ behaves like dt , and those add up to the elapsed time

Brownian motion is differentiable

The third-order terms cancel it

Answer: " $(dW)^2$ behaves like dt , and those add up to the elapsed time". Squared rises of Brownian motion add up to t instead of 0 (quadratic variation), so $\frac{1}{2}f''(dW)^2$ becomes $\frac{1}{2}f'' dt$.

3. Two Brownian motions have correlation $\rho = -0.5$. Over time 0 to 1, the sum of products of their rises approaches...

0

-0.5

0.25

Answer: "-0.5". Their cross-variation is $\rho t = -0.5 \times 1$.